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Acceptance and Kinetic Resolution of α-Methyl-Substituted Aldehydes by Norcoclaurine Synthases

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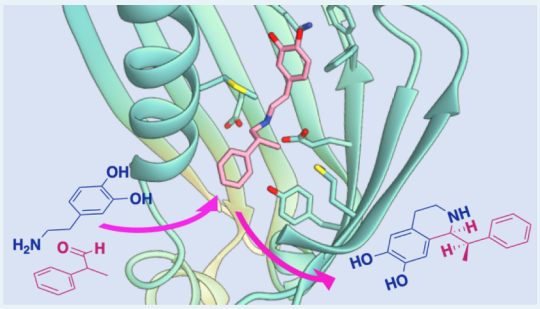
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Supporting Information

ABSTRACT: Norcoclaurine synthase (NCS) catalyzes a stereoselective Pictet−Spengler reaction to give the key intermediate, (*S*)-norcoclaurine in benzylisoquinoline alkaloid biosynthesis. This family of alkaloids contains many bioactive molecules including morphine and berberine. Recently, NCS has been demonstrated to accept a variety of aldehydes and some ketones as substrates, leading to a range of chiral tetrahydroisoquinoline (THIQ) products. Here, we report the unusual acceptance of α-substituted aldehydes, in particular α-methyl-substituted aldehydes, by wild-type *Thalictrum flavum* NCS (Δ33*Tf*NCS) to give THIQ products. Moreover, the kinetic resolution of several α-substituted aldehydes to give THIQs with two defined chiral centers in a single step with high conversions was achieved. Several dopamine analogues were also accepted as substrates, and reactions were amenable to scale up. Active site mutants of *Tf*NCS were then used, which demonstrated the potential to enhance the stereoselectivities in the reaction and improve yields. The rationale for the acceptance of these substrates and improved activity with different mutants has been gained from a co-crystallized structure of Δ33*Tf*NCS with a nonproductive mimic of a reaction intermediate bound in the active site. Finally, molecular dynamics simulations were performed to study the binding of dopamine and an α-substituted aldehyde and provided further insight into the reaction with these substrates.



KEYWORDS: *biocatalysis, norcoclaurine synthase, tetrahydroisoquinoline, Pictet−Spengler, kinetic resolution*

INTRODUCTION

Biocatalysis continues to be viewed as a highly valuable synthetic method with several advantages compared to traditional organic synthetic approaches because of the nonhazardous reagents, ambient reaction conditions, fewer toxic by-products, and higher reaction stereoselectivities. Methods such as enzyme immobilization, novel screening methods, and enzyme mutagenesis are starting to overcome many of the limitations of biocatalysis such as their poor stability, low efficiency, and narrow substrate scope. Enzymes are therefore becoming a more viable, sustainable option for small molecule syntheses.[1]

The in planta role of norcoclaurine synthase (NCS) is the Pictet−Spengler reaction between dopamine **1** and 4-hydroxyphenylacetaldehyde (4-HPAA) to give (*S*)-norcoclaurine as a single enantiomer. Norcoclaurine is the first committed intermediate to the benzylisoquinoline alkaloid (BIA) family of natural products, so the enzyme is present in many BIA-producing plants. NCS isolated from the common meadow rue, *Thalictrum flavum* (*Tf*NCS), is one of the most studied Pictet−Spenglerases.[2−5] Indeed, the X-ray crystallographic structure has been reported together with mechanistic insights, as well as NMR studies,[6] and recently and it has been established to have good substrate scope.[7−9] Notably, NCSs have been shown to accept a variety of aldehydes and ketones in place of 4-HPAA to yield different C-1 substituted tetrahydroisoquinolines (THIQs) in high stereoselectivities as the (*S*)-isomer. Very recently, density functional theory calculations have also been performed to rationalize the stereoselectivity in the NCS reaction to give (*S*)-norcoclaurine.[10]

The THIQ moiety is present in many biologically active molecules, which have been shown to exhibit a variety of beneficial therapeutic activities including antibacterial,[11] antitumor,[12] anticoagulant,[13] and muscle-relaxant properties.[14−16] The THIQ scaffold is therefore a common target in drug discovery and thus there is a strong interest in synthesizing diverse analogues and single isomer products.

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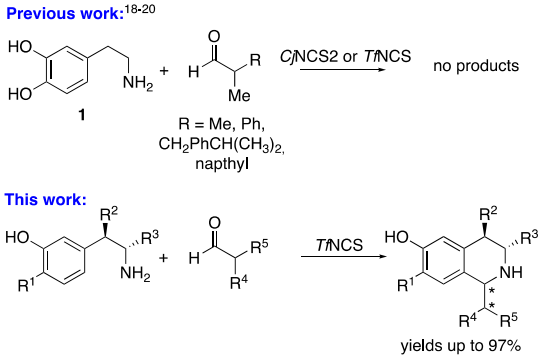
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Accessing single enantiomer THIQ products is challenging using traditional organic synthetic methods and often requires the use of expensive and toxic chiral catalysts, multistep procedures, and often poor enantiomeric excesses (e.e.s) have been reported.[17,18] NCS is proving to be a valuable biocatalytic enzyme and has already been employed in the stereoselective syntheses of THIQs such as trolline, 1,1′-disubstituted THIQs, and spiro-THIQs.[19−23]

NCS has however been reported to show poor tolerance toward α-substituted aldehydes. Previously, when Ruff et al. explored the use of Δ29*Tf*NCS with 2-(1-napthyl)propanal and cyclic carboxaldehydes with dopamine, no Pictet−Spenglerase products were detected, which was rationalized as being due to a lack of conformational space in the active site (Scheme 1).[8,20] The acceptance of 1-naphthalene acetaldehyde

Scheme 1. NCS α-Substituted Aldehyde Substrate Scope Demonstrated in Previous Work and in This Work



but not 2-(1-napthyl)propanal also led to the rationale that more generally α-substituted aldehydes were not well tolerated by NCS. Interestingly, Nishihachijo et al. and Pesnot et al. reported very low levels of dopamine consumption when using the α-methyl-substituted aldehydes isobutanal and 2-phenylpropanal with *Coptis japonica* NCS (*Cj*NCS2), indicative that an NCS catalyzed reaction may have occurred but no products were isolated or characterized.[19,21]

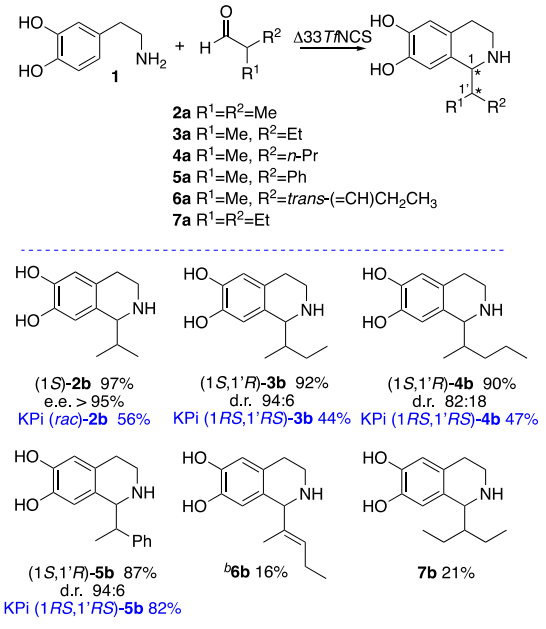
Incorporating α-methyl aldehydes into THIQs is challenging using traditional organic synthetic methods as potentially four different isomers can be formed, unless a single enantiomeric aldehyde is used, which would generate two diastereoisomers. Many single-isomer α-methyl-substituted aldehydes are not commercially available and have been accessed via another available chiral molecule or synthesized using chiral auxiliary methods[24,25] and via asymmetric hydroformylation.[26] However, these methods are step-intensive, often involve the use of toxic reagents, and gaining a high e.e. in the product is challenging. The resulting aldehyde products can also be volatile and prone to oxidation and racemization.[27] Thus, incorporating the single enantiomer aldehyde into the product using nonenzymatic routes is nontrivial and can lead to poor enantiopurity in the product. Accessing the resulting THIQs from reactions between a dopamine analogue and an α-methyl aldehyde is pharmaceutically relevant. Products have been reported to possess bronchodilatory activities and a corresponding isoquinoline has also been shown to have antimalarial activities.[28,29] Corydaline, a 13-methyltetrahydroprotoberberine alkaloid, is also a major constituent of *Corydalis tuber*, which is in clinical trials in Korea for the treatment of functional dyspepsia.[30]

Here, we report the acceptance of several α-substituted aldehydes by wild-type *T. flavum* NCS (Δ33*Tf*NCS) to give THIQ products in high yields. In addition, the kinetic resolution of α-methyl-substituted aldehydes by NCS has been achieved, resulting in the formation of THIQ products with two defined stereocenters. The potential for enzyme engineering has also been explored through observing the reactions of a variety of NCS-active site mutants, resulting in enhanced diastereoisomeric ratios (d.r.s) and increased conversions. A co-crystallized structure of *Tf*NCS with a nonproductive reaction intermediate analogue provided structural insights into the acceptance of these substrates. Molecular dynamics (MD) simulations of enzyme−substrate complexes were also performed to model the binding of dopamine and an α-substituted aldehyde, the (*R*)- and (*S*)-enantiomers of 2-phenylpropanal, to explore the molecular basis of stereoselectivity.

RESULTS AND DISCUSSION

Initial NCS Reactions. Following the reported acceptance of ketones as substrates,[21] with a view to significantly extending the scope of the Pictet−Spenglerase NCS with more sterically challenging aldehydes, a range of α-methyl- and α-ethyl-substituted aldehydes was initially tested as substrates. Reactions were investigated using wild-type NCS (WT-Δ33*Tf*NCS) and dopamine **1** with aldehydes **2a**−**7a** (Scheme 2). Remarkably, in all cases, reactions were observed. Notably,

Scheme 2. *Tf*NCS Reactions with 1 and the α-Methyl- and α-Ethyl-Substituted Aldehydes 2a−7a[a]



[a]Reaction conditions with NCS: dopamine·HCl (10 mM), aldehyde (20 mM), sodium ascorbate (5 mM), Δ33*Tf*NCS (0.2 mg mL^{-1}) in HEPES buffer (100 mM, pH 7.5) with DMSO (10% v/v) at 37 °C, 250 μL scale reactions. Reaction conditions with KPi: dopamine·HCl (10 mM), aldehyde (20 mM), ascorbic acid (5 mM) in potassium phosphate buffer (KPi) (0.3 M, pH 6) with acetonitrile (50% v/v) at 60 °C. Yields were determined by monitoring product formation against standards (by HPLC) unless otherwise indicated; d.r.s for stereoselective reactions were determined by ^{1}H NMR spectroscopy. [b]The conversion was determined by monitoring dopamine consumption (HPLC, Supporting Information Section 9.2).

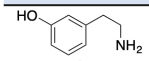
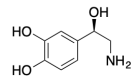
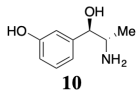
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the α-methyl-substituted aldehydes **2a**−**5a** were readily accepted by *Tf*NCS to give products **2b**−**5b** in high high-performance liquid chromatography (HPLC) yields (87−97%). Aldehyde **6a** was less readily accepted likely because of less rotational freedom of the alkyl chain in the active site and reduced reactivity of the conjugated aldehyde. THIQ **7b** was formed in a lower yield most likely because of steric reasons. Stereoselectivities are discussed below. Analogous reactions to the higher yielding enzyme reactions were also performed without an enzyme, using potassium phosphate (KPi) and acetonitrile (1:1) to give the racemic products **2b**−**5b** in 44−82% yield with no or traces of the ortho-product formed, reflecting the more sterically demanding nature of the α-substituted aldehydes.[31] Racemic products were generated using KPi for use as standards in chiral HPLC analyses. In all cases, reaction yields were lower in reactions using KPi as a catalyst (no NCS present) compared with NCS, highlighting the synthetic value of the NCS-mediated reactions.

To extend the range of accessible THIQ products using α-methyl-substituted aldehydes, other amine analogues were used. Reactions were also observed with *meta*-tyramine **8**, norepinephrine **9**, and metaraminol **10**. Conversions (based on amine consumption are given in Table 1) with aldehydes **2a**−**5a** were lower than those with dopamine and so product formation was confirmed by liquid chromatography−mass spectrometry (LC−MS). The lower conversions are most likely because of the different binding modes of the unnatural dopamine substrates into the active site, leading to a decreased acceptance of the more challenging aldehyde substrates. However, it highlights the more general applicability of using α-methyl-substituted aldehydes as NCS substrates.

Table 1. Conversions When Using α-Substituted Aldehydes and a Variety of Different Amine Analogues[a]

Amine	Aldehyde					
	2a	**3a**	**4a**	**5a**	**6a**	**7a**
 **8**	34%	48%	45%	69%	25%	26%
 **9**	✓	65%	57%	79%	x	x
 **10**	64%	50%	42%	69%	x	x

[a]Reaction conditions: amine (10 mM), aldehyde (20 mM), sodium ascorbate (5 mM), Δ33*Tf*NCS (0.2 mg mL^{-1}) in HEPES buffer (100 mM, pH 7.5) with DMSO (10% v/v) at 37 °C for 18 h, 250 μL scale reactions. Tick mark = product observed but no conversion determined. *x* = no conversion observed (i.e., no product observed). Reactions were performed in triplicate and were consistent within 10% error. Conversions were determined based on amine consumption (HPLC analysis) and confirmed by the detection of the products by LC−MS data (see the Supporting Information).

Reaction Stereoselectivity. For reactions with the α-methylated aldehydes, stereoselectivities were explored for the highest yielding reactions with dopamine **1**. Aldehyde **2a** is achiral: chiral HPLC analysis of **2b** indicated it was formed in >95% e.e., by comparison to the racemic standard prepared using the KPi-mediated reaction (Scheme 2), and following previous reports using *Tf*NCS this was assigned as the (*S*)-isomer.[19,32] THIQs **3b**−**5b** possess two chiral centers, and analysis of the ^{1}H NMR spectra revealed two proton environments for 1-H (ca. 4−5 ppm), indicating that two diastereoisomers were present. As WT-*Tf*NCS typically generates (*S*)-THIQs at C-1, it was rationalized that the stereoselectivity at this position was retained and that a kinetic resolution of the aldehyde had occurred. D.r.s were determined by ^{1}H NMR spectroscopy, and high d.r.s were noted for **3b** and **5b** (94:6). To determine which aldehyde was preferentially accepted, (2*R*)-**3a** and (2*S*)-**3a** were synthesized. Aldehyde **5a** was not prepared because of the fast racemization rates reported at pH 7.5.[33] Aldehyde (2*S*)-**3a** was synthesized from the corresponding alcohol (commercially available in 95% e.e.). The (2*R*)-isomer was prepared using an Evan's chiral auxiliary, subsequent reduction to remove the auxiliary, and oxidation to give (2*R*)-**3a** (Scheme S1). For both isomers, after Swern oxidations the aldehydes were not purified by distillation or chromatographic methods to minimize the risk of racemization: ^{1}H NMR spectroscopic data indicated reasonable levels of purity for both aldehydes, and chiral gas chromatography analysis (Figure S9) indicated that (2*R*)-**3a** was formed in 79% e.e. and (2*S*)-**3a** in 68% e.e.

*Tf*NCS reactions were performed with both (2*R*)-**3a** and also (2*S*)-**3a**; yields were lower than before, most likely because of aldehyde impurities present (see above). HPLC analysis (Scheme 3a) of the reactions under the same conditions indicated that (2*R*)-**3a** was accepted preferentially to the (2*S*)-isomer, confirming that a kinetic resolution was occurring when using the racemate. Indeed, the low levels of product formation with (2*S*)-**3a** (68% e.e.) most likely arose from the small amount of (2*R*)-isomer present (~16%). Following the established selectivity at C-1 for *Tf*NCS and the NMR spectroscopic data, it was concluded that the major isomer formed when using racemic **3a** was (1*S*,1′*R*)-**3b** (d.r. 94:6). It was also rationalized that the same stereoselectivity arises when using **4a** to give (1*S*,1′*R*)-**4b** (d.r. 82:18) as the major isomer out of the four possible isomers.

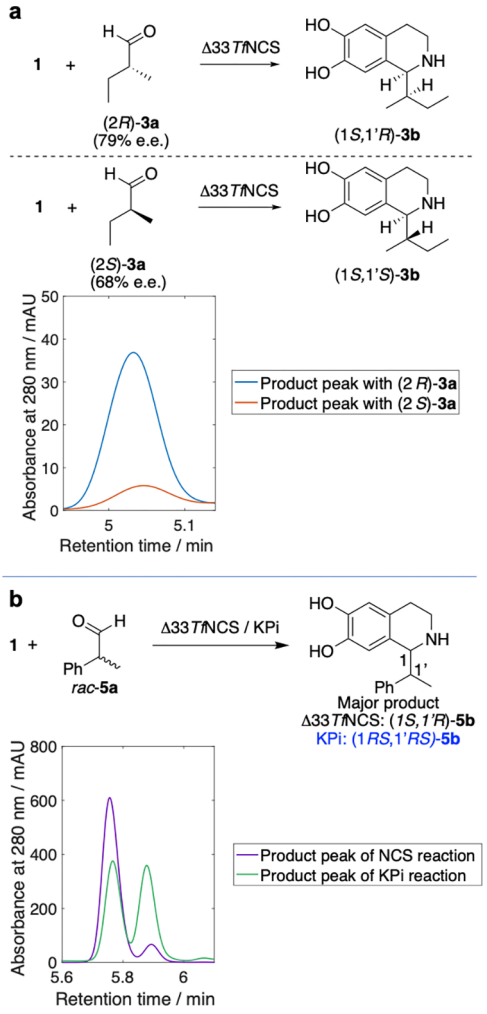
HPLC analysis of **5b** generated using the nonstereoselective KPi-mediated Pictet−Spengler reaction[31] revealed two product peaks corresponding to 4-isomers (two sets of diastereoisomers) in a ratio of 1:1 (Scheme 3b). ^{1}H NMR spectroscopy confirmed that only a trace of the ortho-product was formed. When compared to HPLC data for the enzymatic reaction, it was evident that an excess of one diastereoisomer was formed (ratio ~95:5) (Scheme 3b) in the enzymatic reaction, which was consistent with the ^{1}H NMR data. As arylacetaldehydes react with *Tf*NCS to give the (*S*)-isomer at C-1 in high e.e.s,[19] and following the enantioselectivity data for **3a**, this highlighted that the major product was likely to be (1*S*,1′*R*)-**5b** (d.r. 94:6). Clearly, the enzymatic reaction with WT-*Tf*NCS is an excellent route to such THIQs, as the reactions give higher yields than in the KPi reaction and there is potentially stereochemical control at both chiral centers.

To determine whether the reaction could be readily scaled up, a 0.5 g scale reaction was performed using clarified cell lysate of Δ33*Tf*NCS with **1** and **3a** as the aldehyde substrate. The product was isolated via an extractive workup and the reaction went to 96% yield (HPLC yield, Supporting Information Section 9.2) and an isolated yield of 88% was achieved, consistent with the smaller scale reactions (92% HPLC yield). ^{1}H NMR spectroscopic analysis of the product **3b** determined that the stereoselectivity of the reaction was mostly retained with a d.r. of 87:13 (small scale d.r. 94:6). The

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Scheme 3. (a) Analytical HPLC Analysis (Supporting Information Section 9.2) of Enzymatic Reactions between 1 and Isomers (2*R*)-3a and (2*S*)-3b[a]; (b) Analytical HPLC Analysis of Enzymatic and KPi Reactions between 1 and 5a[b]



[a]Reaction conditions: Dopamine·HCl (10 mM), (*R*)-**3a** (30 mM) or (*S*)-**3a** (30 mM) sodium ascorbate (5 mM), purified Δ33*Tf*NCS (0.2 mg mL^{-1}) in HEPES buffer (100 mM, pH 7.5) with DMSO (10% v/v); The total reaction volume was 250 μL and they were performed at 37 °C with results confirmed in triplicate. [b]Reaction conditions for enzymatic reaction: Dopamine·HCl (10 mM), **5a** (20 mM), sodium ascorbate (5 mM), purified Δ33*Tf*NCS (0.2 mg mL^{-1}) in HEPES buffer (100 mM, pH 7.5) with DMSO (10% v/v) at 37 °C for 18 h, 250 μL reaction volume; Reaction conditions for KPi reaction: Dopamine·HCl (10 mM), **5a** (20 mM), sodium ascorbate (20 mM) in KPi buffer (0.3 M, pH 6) with acetonitrile (50% v/v) at 60 °C for 18 h, 1 mL reaction volume.

reduction in d.r.s is most likely because of a small amount of background phosphate-mediated reaction arising from the use of cell lysate rather than purified enzyme.

Reactions with Enzyme Variants. To investigate the effect of NCS variants on the acceptance of α-methyl aldehydes and products formed, several Δ29*Tf*NCS mutants were used as enzyme lysates, namely, L76V, A79I, A79F, F80L, M97V, M97L, M97V, and Y108F with **1** and **2a**, **3a**, and **5a**. Enzyme variants were chosen based upon docking experiments and previous mutagenesis studies to increase the space in the active site for bulkier substrates.[8,22] First, WT-Δ29*Tf*NCS was used in the reaction and no difference in reactivities or the products formed was found compared with Δ33*Tf*NCS. For all three aldehydes, the mutant A79I showed higher product yields by HPLC compared to WT and the other NCS variants (Figure 1), a characteristic that has also been observed with

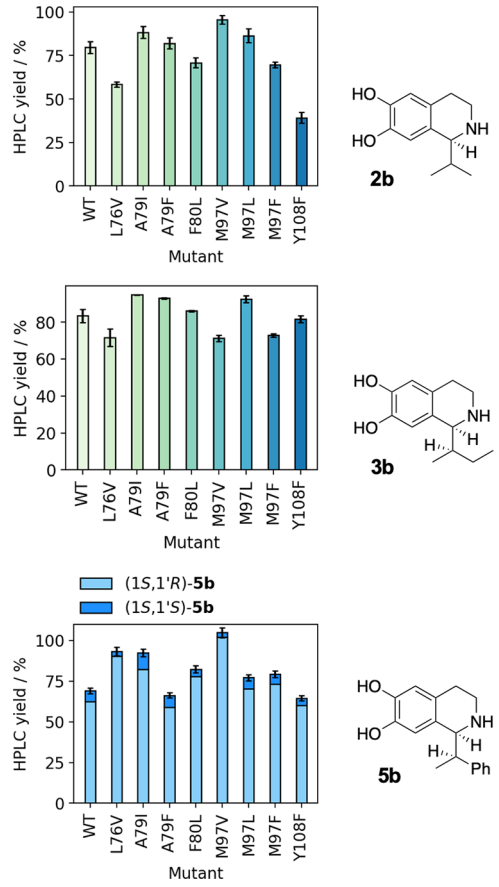


Figure 1. Yields of *Tf*NCS reactions with **1** and **2a** (a), **3a** (b), or **5a** (c) to give products **2b**, **3b**, and **5b**, respectively. Reaction conditions: dopamine·HCl (10 mM), aldehyde (20 mM), sodium ascorbate (5 mM), Δ29*Tf*NCS lysate of variant (25% v/v) in HEPES buffer (100 mM, pH 7.5) with DMSO (10% v/v) at 37 °C for 18 h, 200 μL scale reactions; yields were determined by monitoring product formation against standards by HPLC (Supporting Information Section 9.2). d.r.s of the major (1*S*,1′*R*) product to the minor (1*S*,1′*S*) product for each reaction product of **5b** are WT, 90:10; L76V, 97:3; A79I, 89:11; A79F, 89:11; F80L, 95:5; M97V, 97:3; M97L 91:9; M97F, 92:8; Y108F, 93:7. Reactions were performed in triplicate and error bars are the standard deviations.

ketones but to a greater extent.[22] This mutation has been rationalized to increase conversions of hydrophobic substrates because of enhanced hydrophobicity in the active site resulting in increased substrate affinity. For the two chiral aldehydes **3a** and **5a**, M97V showed increased yields compared to the wild-type NCS. M97 mutants have been used with ketone substrates where decreased yields were noted with a reduction in steric bulk at this position, that is, poorer conversions with M97V compared with the wild-type.[22] Interestingly, the opposite trend was observed here with aldehydes **2a** and **5a** as substrates.

HPLC analysis of the reactions between **1** and **5a** showed that different active site mutants of *Tf*NCS gave different d.r.s of the products. A ratio of 90:10 was observed for wild-type

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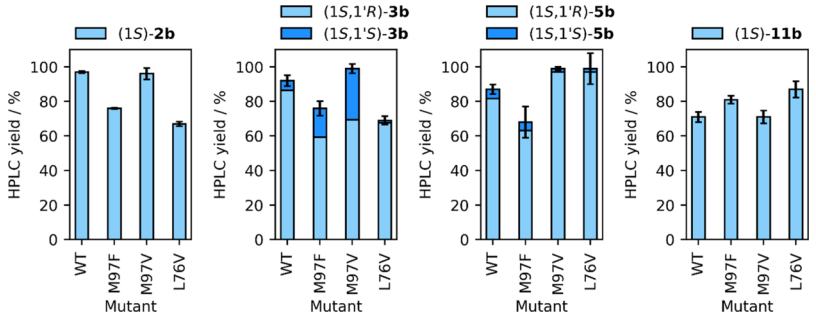
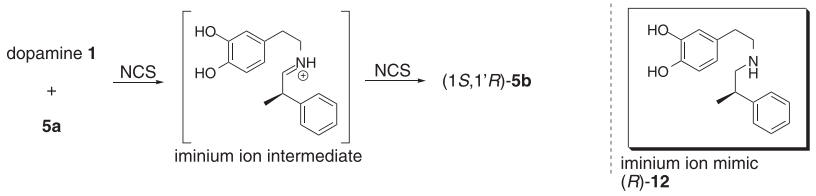


Figure 2. Yields and e.e.s of reactions of products of enzymatic reactions using purified *Tf*NCS active site mutants. Reaction conditions: dopamine·HCl (10 mM), aldehyde (20 mM), sodium ascorbate (5 mM), purified *Tf*NCS (0.2 mg mL^{-1}) in HEPES buffer (100 mM, pH 7.5) with DMSO (10% v/v) at 37 °C for 18 h; yields were determined by product formation against standards using HPLC (Supporting Information Section 9.2). Reactions were performed on a 250 μL scale for **2b**, **5b**, and **11b**. Reactions were performed on a 1 mL scale for **3b**. e.e.s were determined by chiral analytical HPLC for **2b** and **11b**. d.r.s were determined by ^{1}H NMR spectroscopy for **3b**. The d.r. was determined by analytical HPLC for **5b**.

Scheme 4. Design of Mimic (*R*)-12 of the Reaction between 1 and 5a, for Co-Crystallization Based upon the Ligand Previously Co-Crystallized with Δ33*Tf*NCS (PDB: 5NON)



NCS lysate, lower than with purified NCS (94:6), most likely because of the increased background phosphate reaction when using lysate. The two most promising active site mutants observed were L76V and M97V, where product d.r.s of 97:3 were observed. To fully explore the change in reaction selectivity of these two mutants, both mutant enzymes (M97V and L76V) were purified and reactions performed between **1** and **2a**, **3a**, **5a** and phenylacetaldehyde (**11a**). The mutant M97F was also explored to provide a comparison with the M97V mutant (Figure 2).

Aldehyde **11a** was used as a substrate with all three mutants to help determine whether the variants were affecting the stereochemistry of the product at the C-1 position. As previously reported, the reaction with wild-type *Tf*NCS gave a high e.e. in the product **11b**,[19] as did all the NCS variants tested here, implying that there was no effect on the stereochemistry at the C-1 position. For all reactions with **2a** as the aldehyde substrate, a high e.e. (>95%) in the product **2b** was also retained. The enantiopurity of **2b** also demonstrated that the variants were not affecting the stereochemistry at the C-1 position of the products with α-methyl aldehydes as substrates. Yields with M97F and L76V and **2a** as the aldehyde substrate were however slightly decreased compared to the wild-type.

Most notably, for all three α-methyl aldehydes tested as substrates, the products of reactions with Δ29*Tf*NCSL76V showed the same or improved stereoselectivities compared with the wild-type enzyme. There was a lowering in d.r. for all products in M97F reactions with the chiral α-methyl aldehydes, suggesting that this mutant is affecting the stereochemistry of the C-1′ position. With the bulky, chiral aldehyde **5a** however, the d.r. of the product **5b** was improved compared with the wild-type for both M97V and L76V mutants, and both mutants gave the product in almost quantitative yields. For both mutants, there was also little change in d.r. between using a purified enzyme here compared with enzymatic lysate (Figure 1), highlighting that purified enzyme is not essential for gaining a high d.r. in the product in these reactions.

Co-Crystallized Structure with an α-Methyl Reaction Intermediate Analogue. To rationalize the acceptance of α-methyl aldehydes by NCS, a crystal structure of Δ33*Tf*NCS with a nonproductive mimic of a reaction intermediate bound in the active site was obtained. This was based upon previously published work in our group, wherein the "dopamine-first" mechanism of *Tf*NCS was confirmed via a co-crystallized structure of Δ33*Tf*NCS with a nonproductive mimic of the reaction.[9] A nonproductive, secondary amine mimic (*R*)-**12** of the iminium ion intermediate of the reaction between **1** and (*R*)-**5a** was synthesized using an analogous method to the previously reported co-crystallized mimic, starting with 3,4-bis(benzoyloxy)dopamine followed by amide coupling, reduction to give the amine, and subsequent deprotection to give the mimic in 14% yield. The enantiopurity of the mimic was confirmed by chiral HPLC (method 2, Supporting Information Section 9.3) as >99% e.e., using a racemic standard **12**, synthesized in an analogous manner. A single enantiomer mimic (*R*)-**12** (Scheme 4) was used, as the (*R*)-enantiomer of α-methyl aldehydes was preferentially accepted as substrates by the NCS as described above. The "aldehyde-part" of the mimic was chosen to be (*R*)-**5a** as the phenyl group would provide extra electron density and aid identification of the ligand in the active site. Indeed, initial attempts to co-crystallize an unproductive mimic of the reaction between dopamine and racemic **3a** resulted in multiple conformations of the mimic in the active site, so the ligand could not be clearly built into the density.

A co-crystallized structure of Δ33*Tf*NCS with mimic (*R*)-**12** in the active site was gained at 1.81 Å with a single copy in the asymmetric unit (PDB entry: 6RP3, Table S1). Compared to

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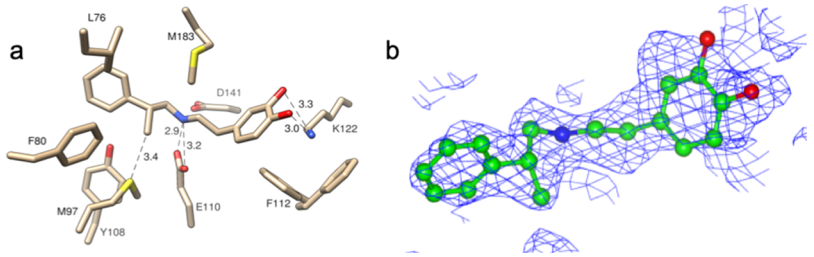


Figure 3. (a) Location of mimic (*R*)-**12** in the active site. Dotted lines show key distances or hydrogen-bonding interactions. Associated numbers are distances given in Å. (b) Electron density of (*R*)-**12** in the active site at contour level 0.81 RMSD.

the previous apo structure of Δ33*Tf*NCS (PDB: 5N8Q), the overall fold of NCS was retained (RMSD C_α = 0.587). The only notable difference in active side residue conformations was with Glu110 and Met97. The mimic density in the active site was less convincing than with the previously co-crystallized structure (PDB: 5NON), suggesting partial occupancy and other, minor multiple conformations of the ligand. A Polder[34] map of the ligand density showed similar density features, with some uncertainty in the placement of the dopamine ring. However, the "aldehyde-end" of the mimic could be observed relatively clearly and so this was used to help understand why these substrates are accepted and why activity varies with single-point mutants.

In this structure (Figure 3), the ligand is placed with both hydroxyl groups of the catechol hydrogen-bonded to Lys122 (with distances of 3.28 and 3.02 Å, respectively) in a linear configuration, likely reflecting a pre-cyclization conformation. Previous studies involving the active site mutagenesis of key residues in NCS have shown that Lys122 is essential for catalytic activity.[7,8] The hydrogen-bonding distances are further away than in 5NON and the mimic appears to be further buried into the active site. A hydrogen-bonding interaction can also be observed between the secondary amine of the mimic and Glu110, which is in a rare conformation, observed in only 1% of glutamic acid residues in the PDB, whereas in the previous structure, the amine was bound to both Glu110 and Asp141. This interaction is consistent with the "dopamine-first" mechanism (detailed in the Supporting Information Section 4) and we are likely observing another "snapshot" of the mechanistic process. Phe112 is present in two different conformations in the novel structure (defined at 1:1 occupancy), as in the first holo structure of *Tf*NCS (PDB: 2VQ5).[7] In subunit B of 2VQ5, the phenylalanine conformation is uncommon, being present in only 1% of phenylalanine residues in the PDB. In structure 5NON, Phe112 is predominately in the conformation predicted by MD simulations and in subunit A of 2VQ5.[8] It is not known whether this unusual phenylalanine conformation is relevant to the NCS mechanism, perhaps suggesting an inhibitory mode with the residue acting as a "gatekeeper", or if it is simply a crystallographic artifact.

The density of the "dopamine-end" of the mimic was less clear so the ligand was built in with both hydroxyl groups hydrogen-bonded to Lys122, reflecting a likely mixture between the two potential conformations, productive and inhibitory. The phenyl group of the "aldehyde-end" of the mimic sits close to Leu76. Mutation of this residue to valine resulted in improved stereoselectivity for NCS reactions with α-methyl aldehydes. The "aldehyde-end" of the mimic can be observed clearly and the methyl group of the mimic appears to sit close to Met97, forcing the residue into a previously unseen conformation for this residue in NCS structures. Reducing bulk at this position by mutation to a valine residue lowers d.r.s for the reaction with aldehyde **3a**, thus providing extra space for an ethyl group and so both enantiomers of the aldehyde are more readily able to react with dopamine in the active site. However, this does not provide enough extra space for the phenyl group of **5a** and so one enantiomer is accepted preferentially. An increased d.r. in the product **5b** compared to the wild-type is likely because of improved binding of the aldehyde from this extra space provided and also leads to increased reaction yields. Conversely, the mutant M97F reduced yields and d.r.s in all products, likely because of lack of space in this region of the active site. The mutant L76V improved d.r.s with both chiral aldehydes, likely altering the shape of the active site cavity so that binding of the (*R*)-enantiomer is preferential.

Molecular Dynamics Simulations and Mechanistic Insights. With the aim of further understanding the preference for the (*R*)-enantiomer of the α-methyl aldehydes used, MD simulations were performed. Four different starting structures with dopamine and 2-phenylpropanal bound in different positions were defined, using chain A of PDB:5NON with the ligand removed (Figure 4). In starting structures A and B, **1** was placed inside the active site, and the (*R*)- and (*S*)-enantiomers of **5a** (structures A and B, respectively) were placed at the active site entrance. In starting structures C and D, the two **5a** enantiomers were placed inside the active site ((*R*)- and (*S*)-enantiomer, respectively), and **1** at the active site entrance.

Out of the 60 simulations launched, 58 were completed successfully and two simulations with starting structure B aborted during minimization, resulting in a total of 18.45 μs simulation time. Analyses of near attack complexes (NACs, conformers with d_{NAC} < 4.0 Å, see Supporting Information Section 6 for details) are summarized in Figure 5. Selected atoms of the two substrates were used for the calculation of d_{NAC} (see Supporting Information eq 1 for details). The 1858 conformers considered to be near attack complexes amount to 0.04% of all conformers, illustrating that the productive binding of substrates is a very rare event. Starting structure A (**1** inside the active site and (*R*)-**5a** at the protein surface) yielded most NACs with d_{NAC} < 4.0 Å (1745 conformers). Starting structures with d_{NAC} < 4.0 Å, B and C, only yielded 90 and 23 conformers, respectively. Starting structure D did not result in any conformers with d_{NAC} < 4.0 Å. The number of NACs with d_{NAC} 4.0−4.5 Å are also shown for illustrative purposes and a similar ratio of conformers across the starting structures was observed.

On the basis of analysis of all NACs with d_{NAC} < 4.0 Å (Figure 5), it is possible to conclude that the (*R*)-enantiomer of **5a** is preferred over the (*S*)-enantiomer with a ratio 95:5

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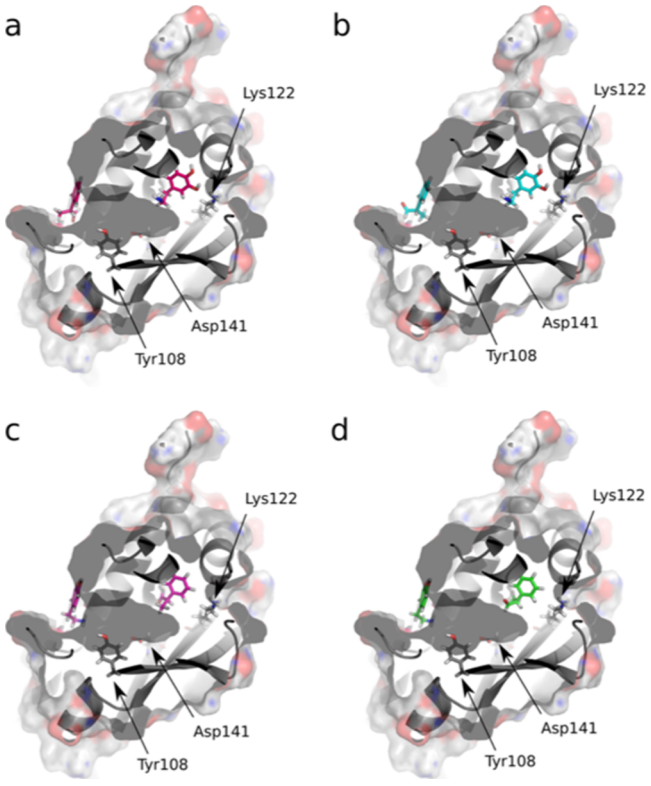


Figure 4. Four different starting structures of substrates **1** and **5a** bound to NCS. NCS is shown as a cartoon and surface representation, allowing for identification of the active site binding pocket located as a void volume, approximately above Tyr108 and Asp141, delimited by Lys122 (right) and Tyr108 (left). Oxygen, nitrogen, and hydrogen atoms are colored in red, blue, and white, respectively. NCS is colored in gray and different colors of the bound substrates indicate the different starting structures. Dopamine was placed inside the active site, and the (*R*)- and (*S*)-enantiomers of **5a** (A and B, respectively) were placed at the protein surface close to the active site entrance. The aldehyde **5a** enantiomer was placed inside the active site ((*R*)- and (*S*)-enantiomer, respectively), and dopamine **1** at the protein surface at the active site entrance (C and D, respectively). In all subsequent figures, the colors red, blue, magenta, and green are used to indicate simulations starting from structures A, B, C, and D, respectively.

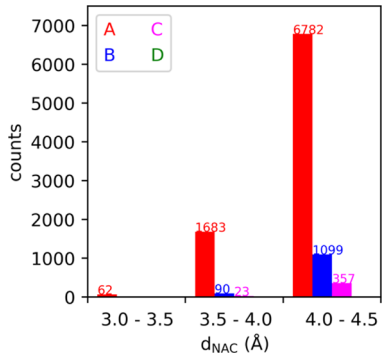


Figure 5. Near attack conformation distances below 4.5 Å for the four different starting conformations A, B, C, and D (Figure 4). The number of frames with NACs between 4.0 and 4.5 Å is shown for illustration purposes.

conformers (1745:90 conformers), resulting in the preferred production of the (1*S*,1′*R*)-product (**5b**). This preference agrees with the experimentally observed selectivity of NCS. D.r.s for these substrates were 94:6 (Scheme 2), between the (1*S*,1′*R*) and the (1*S*,1′*S*) products when racemic aldehyde was used. NAC_A therefore is most likely the NAC leading to the formation of the (1*S*,1′*R*)-product. The catalytic activity of NCS with **5a** and dopamine **1** is rather low, indicated by the need for high substrate loading (10 mM), which is in agreement with our observation in MD simulations, that productive substrate binding is a rare event, as only 0.04% of conformers show an NAC with d_{NAC} < 4.0 Å.

Visual analysis of the NACs led to the choice of representative conformers with d_{NAC} < 4.0 Å (Figure 6). Inspection of the substrate orientations in the active site showed that substrates showed consistent binding orientations in conformers originating from starting structure A with d_{NAC} < 4.0 Å (Figure 6a, NAC_A). Two different, distinct binding orientations were visible for conformers originating from starting structure B with d_{NAC} < 4.0 Å (Figure 6b,c, NAC_{B1}, NAC_{B2}). Another binding orientation was found for conformers originating from starting structure C with d_{NAC} < 4.0 Å (Figure 6d, NAC_C). Examination of the NACs showed that in NAC_A, dopamine was oriented with the amine toward the amine of Lys122, with the hydroxy groups pointing further into the active site pocket, consistent with the dopamine-first mechanism. (*R*)-**5a** was located closer to Tyr108, with its phenyl ring oriented toward Tyr108 and its aldehyde group pointing toward the amine group of dopamine. Aldehyde (*R*)-**5a** was therefore located closer to the exit of the active site. For an illustration, see the movie (NACA.avi) in the Supporting Information. From analysis of the NACs with (*R*)- and (*S*)-**5a**, NAC_A was considered to be the most likely NAC leading to the formation of the (1*S*,1′*R*)-product, so inspection of representative conformers for the other NACs will be considered briefly. NAC_{B1} was similar to NAC_A: (*S*)-**5a** was located closer to Tyr108, with its phenyl ring oriented toward Tyr108. Dopamine was located close to Lys122. In NAC_{B2}, dopamine was located below and slightly to the left of (*S*)-**5a**. The amine of dopamine pointed away from Tyr108 and the two hydroxyl groups pointed toward Asp141. In NAC_C, **1** was located close to Tyr108, with its amine group oriented toward the hydroxyl group of Tyr108. All molecules were allowed to move freely, so it was possible that both substrates unbind from NCS and interact only with each other and the solvent. This was only observed in nine conformers with d_{NAC} < 4.0 Å. Root mean square function (RMSF) data showed a trend toward slightly higher RMSF values for simulations with starting structures A and B (Figure S6). This trend was stronger when only considering NAC with d_{NAC} < 4.0 Å (Figure S7). It is unclear whether this slightly increased flexibility impacts the mechanism of NCS.

Two conflicting mechanisms have been proposed for aldehyde activation and loss of water to allow formation of the iminium ion intermediate.[8,9] Briefly, in the "aldehyde first" mechanism, aldehyde activation is proposed to involve Lys122, with Tyr108 having a role in catechol deprotonation. However, as experimentally, bulky aldehyde substrates are converted to THIQs, subsequent in silico modeling and a crystal structure with an imine intermediate analogue led to the proposed "dopamine-first" mechanism.[8,9] In this mechanism, Glu110 is suggested to play the major role (together with Asp141 and Tyr108) in aldehyde activation and imine formation, with Lys122 deprotonating the catechol.

In the reaction between **1** and (*R*)-**5a**, starting structure A where dopamine is positioned inside the active site leads to the

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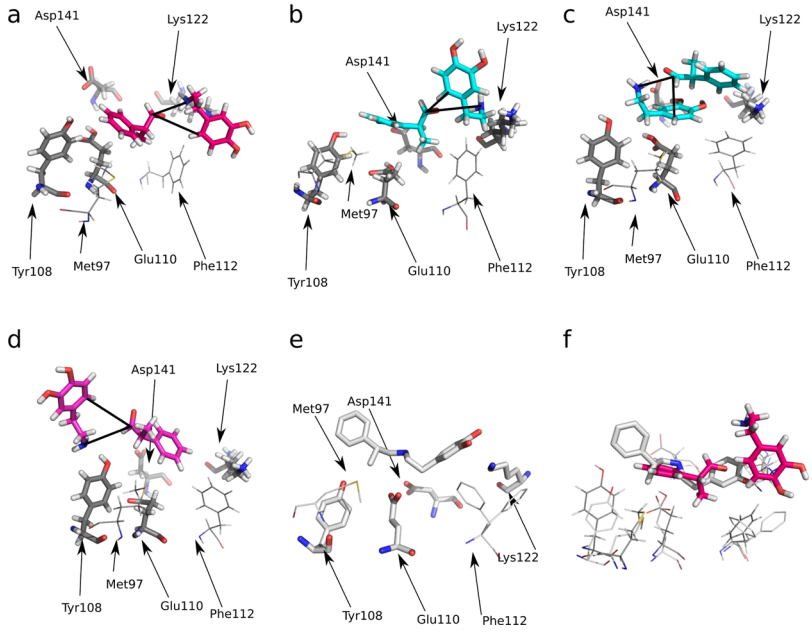


Figure 6. a–d are representative near attack complexes (NAC_A, NAC_{B1}, NAC_{B2}, and NAC_C) of simulations with starting structures A, B, and C. Black, thick lines indicate the distances between atoms used to calculate d_{NAC}. (e) shows the iminium ion mimic from PDB 6RP3 from the same viewpoint as the representative NACs. (f) shows NAC_A aligned with the iminium ion mimic from PDB 6RP3. Note that (e) shows parts of a crystal structure, thus missing hydrogen atoms, and that (f) shows the crystal structure (in white) aligned with NAC_A, which contains hydrogen atoms that were added for the simulations.

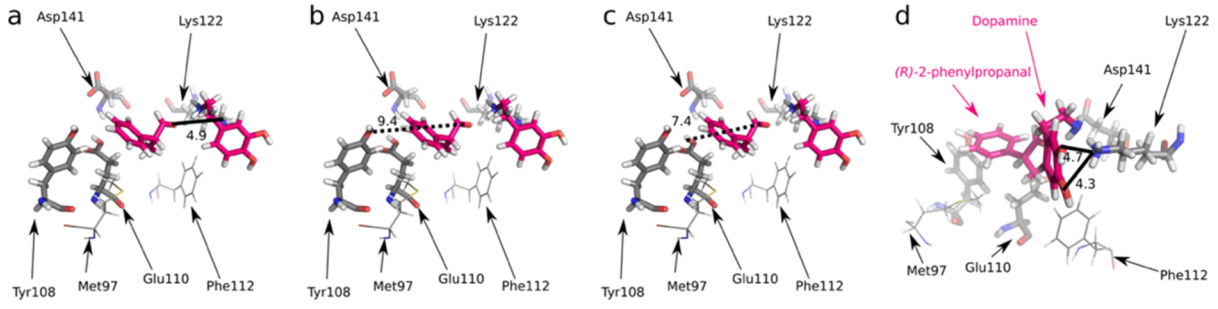


Figure 7. Distances of the aldehyde functional group to key catalytic residues in the most prevalent near attack complexes, originating from starting structure A, NAC_A. Distance from the carbon of the aldehyde functional group to the (a) nitrogen on Lys122, (b) phenolic oxygen of Tyr108, (c) oxygen of Glu110. (d) Distances from the dopamine catechol oxygens to the nitrogen of Lys122.

most NACs (NAC_A). This indicates that the reaction is much more likely to happen if dopamine is bound to NCS first, followed by the aldehyde. Distances in NAC_A between substrate and the proposed catalytic residues were measured as an indicator for catalytic activity (Figure 7).

For NAC_A, the distances between the substrate aldehyde carbon and Lys122 nitrogen and Tyr108 phenolic oxygen were 4.9 and 9.4 Å, respectively (Figure 7a,b). In addition, the distance between the substrate aldehyde carbon and Glu110 was 7.4 Å (Figure 7c). Assuming the dopamine-first mechanism, this suggests that upon aminol formation, a movement of the substrate aldehyde toward Glu110 is required to facilitate the loss of water.[8] The orientation of the meta-hydroxyl group of dopamine is crucial for NCS activity,[7] which is supported by the observed distances between the hydroxyl groups of dopamine to the Lys122 nitrogen (4.7 and 4.3 Å, for meta and para hydroxyl groups, respectively, Figure 7d). Distances of the dopamine amine to Asp141 and Glu110 were too far to allow hydrogen bonding (10.1 and 9.8 Å, respectively, data not shown). Assuming the dopamine-first mechanism, this suggests that upon aminol formation, a movement of the substrate toward Glu110 is required to facilitate the imine formation and cyclization. Indeed, the previous NMR study highlighted structural changes upon substrate complexation.[6] The relative orientation of the substrates to each other in NAC_A is identical to the iminium ion mimic in the crystal structure, although the position of the substrates is slightly shifted (Figure 6e,f). The observed differences could indicate that upon formation of the aminol or the iminium ion intermediates, a slightly different binding interaction in the active site arises, compared to the individual substrates. In NAC_{B1}, NAC_{B2}, and NAC_C the orientation and position of substrates is different from the crystal structure.

MD simulations were performed in a workflow, allowing repeatable simulation of the different starting structures in 15 replicates, as described in more detail in the supporting information. We would like to highlight the fact that due to our workflow, we were able to conveniently perform and analyze in total 60 molecular simulations (4 starting structures, 5 replicates of 100 ns each, and 10 replicates of 400 ns each). Extensive sampling by long simulations and multiple replicates is crucial for obtaining sufficient sampling of rare events like

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productive binding of substrates to the active site. Overall, the simulations were useful to rationalize the enantioselectivity observed and indicate initial positioning of the substrates in the active site.

In summary, this work shows that α-methyl-substituted aldehydes are accepted as substrates by wild-type *Tf*NCS using both dopamine and a variety of analogues as the amine substrate. Products were generated in high yields (up to 97%) with high e.e. (>99%), or d.r.s (up to 94:6) with chiral α-methyl-substituted aldehydes, providing a sustainable, facile route to these compounds. For chiral α-methyl-substituted aldehydes, the THIQ products were generated with two defined chiral centers, with (1*S*,1′*R*)-stereochemistry as determined by the synthesis of two enantiomeric aldehydes. Active site mutants were capable of improving the d.r.s in the products (up to 98:2) with chiral α-methyl-substituted aldehydes and improved the reaction yields with several aldehydes. The enzymatic reaction was confirmed to occur via a mechanism that involves dopamine binding first to the active site, from a co-crystallized structure of Δ33*Tf*NCS with a reaction intermediate mimic in the active site. MD simulations were performed to investigate the binding of dopamine and the (*R*)- and (*S*)-enantiomers of 2-phenylpropanal, into the NCS active site. The simulations provided supporting data that a kinetic resolution of the α-methyl-substituted aldehyde was occurring and that the (*R*)-enantiomer of aldehyde was accepted preferentially.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.9b02699.

Experimental methods, supporting figures and tables, and chemical characterization (PDF)

NCS−MD simulation workflow (ZIP)

NACA movie (AVI)

Accession Codes

The crystal structure reported has been deposited in the Protein Data Bank as entry 6RP3.

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Author Contributions

The paper was written through contributions of all the authors. R.R. performed chemical syntheses, chemical characterization, enzymatic assays, and created the figures. R.R. and A.S conducted protein purifications and crystallization trials. A.S. collected X-ray diffraction data, which were processed and analyzed by N.K., A.S., and R.R. G.G. performed MD simulations and their analysis and created the figures. D.M.-S. expressed the *Tf*NCS mutants. The project was supervised by J.M.W, N.H.K, J.P., and H.C.H. The paper was written by H.C.H, J.P, R.R., and G.G. All the authors have given approval to the final version of the paper.

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

NCS, norcoclaurine synthase; BIA, benzylisoquinoline alkaloid; THIQ, tetrahydroisoquinoline; *Tf*, *Thalictrum flavum*; 4-HPAA, 4-hydroxyphenylacetaldehyde; NMR, nuclear magnetic resonance; e.e., enantiomeric excess; WT, wild-type; KPi, inorganic phosphate; DMSO, dimethylsulfoxide; d.r., diastereomeric ratio; HPLC, high-performance liquid chromatography; LC−MS, liquid chromatography−mass spectrometry; GC, gas chromatography; MD, molecular dynamics; NACs, near attack complexes; RMSF, root mean square function

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